

11.4 Making conclusions

1 Put the following in order to show the steps taken to form a conclusion:

- Use data clearly to support your argument.
- Write a clear statement about the data that supports how strongly the hypothesis was found to be either true or false.
- Acknowledge any unusual results and if possible, explain the reason for them.
- Revisit the hypothesis to determine what you were testing.

2 Read the following scenario and answer Questions a to e below:

A group of geography students conducted fieldwork in a city centre to investigate the relationship between pedestrian footfall and proximity to public transport hubs. They used a combination of pedestrian counts, questionnaires and mapping to collect their data.

The data they collected included:

Pedestrian counts at three locations (A, B and C):

- Location A (next to the main train station): 350 people/hour
- Location B (500 m from the train station): 220 people/hour
- Location C (1 km from the train station): 120 people/hour

Questionnaire results:

- 80% of respondents near Location A said they used public transport to access the city centre.
 - 50% of respondents near Location B used public transport.
 - 30% of respondents near Location C used public transport.
- a** What patterns can be observed in the pedestrian count data?
 - b** What patterns can be observed in the questionnaire results?
 - c** Based on the data provided, suggest one possible conclusion about the relationship between the pedestrian count and proximity to public transport hubs.
 - d** Identify one limitation of the data collection methods used in this study.
 - e** Suggest an additional data collection method that could strengthen the students' conclusions.

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2 A student is measuring the velocity of a river using a float method. Their results are shown in Table 11.3.

Site	Velocity (m/s)
1	0.2
2	0.3
3	0.5
4	0.6
5	0.8

Table 11.3: River velocity at five different sites

Estimate how long it would take for the float to travel 1 metre at Site 3. Choose the closest estimate from the options below:

- a** 0.5 seconds
- b** 1 second
- c** 2 seconds
- d** 5 seconds

[1]

~~4~~ A student found the following hypothesis to be true: 'Beach material gets larger towards the top of the beach.' Their results can be found in Table 11.4 below.

Site	Size of beach material (mm)
1 (close to cliff)	40
2	32
3	—
4	14
5 (close to sea)	8

Table 11.4: Beach material size at five different sites

Estimate the size of beach material at Site 3 from the options listed:

- a** 37 mm
- b** 22 mm
- c** 9 mm
- d** 7 mm

[1]

5 A student measured the height of five plants at different sites along a sand dune transect. Their results for Site 5 are shown in Table 11.5.

Plant number	Height (cm)
1	10
2	13
3	11
4	14
5	12

Table 11.5: Plant heights at Site 5

Estimate the average plant height at this site. Choose from:

- a** 10 cm
- b** 12 cm
- c** 14 cm
- d** 16 cm

[1]

SELF-ASSESSMENT CHECKLIST

Let's revisit the knowledge focus and exam skills focus for this topic.

Decide how confident you are with each statement.

Now I can	Show it	Needs more work	Almost there	Confident to move on
Describe the steps taken in the route to geographical enquiry	What steps must be taken when carrying out a geographical enquiry, and what order do they go in?			
Form hypotheses	What makes a good hypothesis, and can you write one?			
Decide on suitable data collection techniques	How many data collection techniques can you name?			
Plan how to stay safe whilst out in the field	List some risks that could be encountered when carrying			

Now I can	Show it	Needs more work	Almost there	Confident to move on
	out fieldwork and suggest how to limit them.			
Present and analyse data using a suitable technique	List some techniques used to present and analyse quantitative and qualitative data.			
Make accurate conclusions	What steps must be taken when making conclusions?			
Understand the command word 'devise'	What things need to be considered when devising a questionnaire?			
Understand the command word 'estimate'	Which feature of a map allows you to estimate distance or area?			
Understand the command word 'suggest'	Suggest a method to collect data about the width and depth of a local river.			

